

IEORE4004

Optimization Models and Methods

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Agenda

- Locational marginal pricing
- Case Introduction
- Model Formulation
- Price Signals & Scenarios
- Conclusions

Locational Marginal Pricing

Optimization modelling is often used to improve the efficiency of supply chains.

Where there is *vertical integration*¹, an organization can model and optimize each part of the chain that they control.

But how can they coordinate efficiently with independent suppliers and contractors?

¹**Vertical Integration:** a business strategy where a company owns or controls multiple stages of its supply chain, from raw material production to final retail distribution.

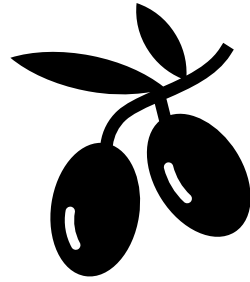
Locational Marginal Pricing

One approach is to incentivize behaviors via **price signals**.

E.g., some wholesale electricity markets use a linear program to find the least-cost suppliers to meet demand given various constraints¹

- The model's dual values are used as prices for buying and selling of electricity and related services.
- If the prices are calculated depending on geographic location, then they are called Locational Marginal Prices (LMP): indicating how the value of electricity varies by location.

[1] <https://lmpmarketdesign.com/lmp.php>



Case Study | Locational Marginal Pricing of Olives

The Current Situation



We operate a chain of restaurants in six cities located on the island nation.

To make a key product, **olive oil**, we purchase olives from suppliers on the island and contractors transport them to our restaurants.

The Current Situation

Each supplier sells olives at a specified price per kg and has a maximum supply capacity (kg/week).

We currently purchase from the cheapest suppliers, at Nodes 1, 2, and 3. Supply is currently not available from the more expensive suppliers at Nodes 4, 5, and 6.

Supplier	\$/kg	Max kg
Node 1	0.80	100,000
Node 2	1.05	60,000
Node 3	0.90	25,000
Node 4	1.10	0
Node 5	1.12	0
Node 6	1.15	0

The Current Situation



The cheapest suppliers are in the north and west of the island

Our largest demand center is in the southeast.

Olives are transported from suppliers in the northwest towards consumers in the southeast, supplying our locations along the way.

The Current Situation

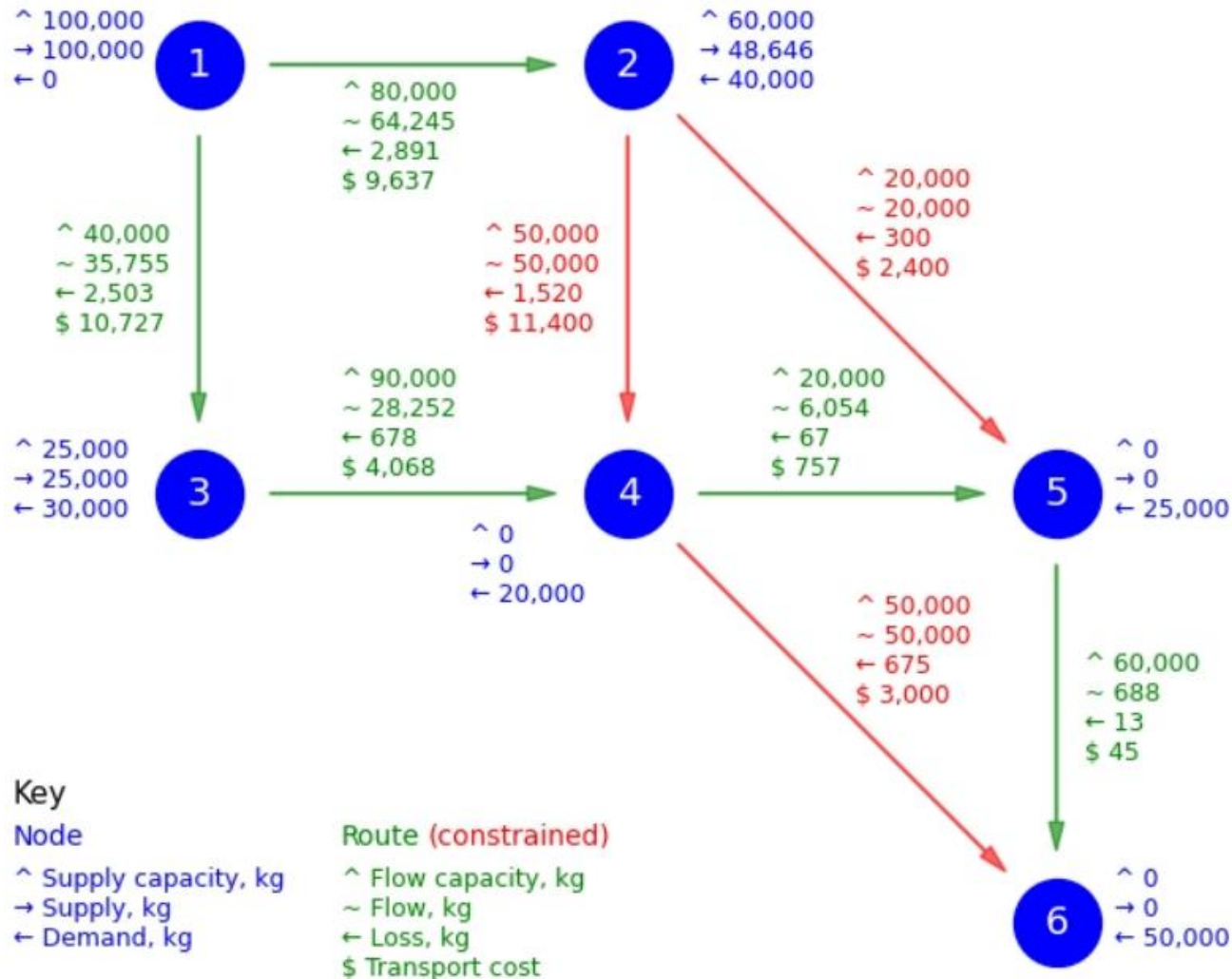
Nodes	Max kg	km	\$/100km	Loss/100km
1-2	80,000	150	0.10	3.0%
1-3	40,000	200	0.15	3.5%
2-4	50,000	95	0.24	3.2%
2-5	20,000	100	0.12	1.5%
3-4	90,000	120	0.12	2.0%
4-5	20,000	50	0.25	2.2%
4-6	50,000	75	0.08	1.8%
5-6	60,000	60	0.11	3.1%

The transportation network characteristics are shown in the Table.

Note that some of the olives will spoil, becoming unusable, during transportation.

Our total cost for olives is **\$ 195,612.49/week**. Around 80% of the delivered cost is for the olives, with the other 20% being transportation costs.

The Current Situation



We represent the locations of the suppliers and our restaurants as nodes, with transportation links between some nodes.

Notes:

- Transportation can occur in either direction on a link, but it makes sense to have transportation in only one direction at a time.

Note: The distance between nodes is not drawn to scale.

Current Problems

Our current supply situation has several issues:

- Our business operates on very thin profit margins, so anything we can do to reduce the current average delivered olive cost of \$1.185/kg (\$195,612.49/week for 165,000 kg/week) would be valuable.
- Two of our three suppliers are operating at their supply capacity, so we have only one supplier (at Node 2) who can meet any increase in our demand.
- The transportation network has significant constraints, which limit our ability to meet our demand efficiently. Specifically, our Supplier at Node 2 is limited by transport capacity constraints.

Current Problems

Our current supply situation has several issues:

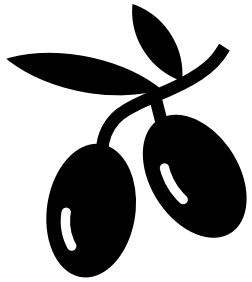
- The overall spoilage rate of 5.0% is high relative to the industry benchmark of 3.0%.
- Spoilage is costing us \$10,245/week (8,646 kg loss * \$1.185/kg).
- Our transportation costs seem excessive, ~20% of total delivered cost.

Additionally, we are one of several restaurant chains, plus there are other substantial olive purchasers (including supermarkets).

Therefore, we have little or no power when negotiating with the olive suppliers and transportation contractors.

? Question

How can we solve our cost and capacity issues?



How can we incentivize the olive suppliers and transportation contractors to offer us a better deal?

Locational Marginal Pricing As A Price Signal

One way to create better incentives for the olive suppliers and transportation contractors is to tell them how much olives are worth to us.

That is, send them a market price signal.

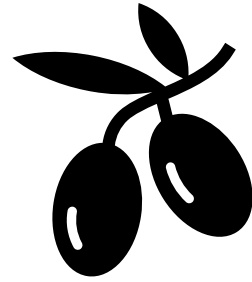
Since our situation covers several locations around the island, the price signal may vary by location.

Locational Marginal Pricing As A Price Signal

- The purpose of LMP is to signal locational value to incentivize efficient behavior.
- Though note that the incentives apply to both suppliers and us.
- If the cost of delivered olives at a specific location is too high, then we should consider whether continuing to have a restaurant at the location is viable, but that's a longer-term decision for another time.

? Question

For addressing the issues in our immediate situation, how do we calculate the locational marginal prices?



Linear Program Formulation

Model Formulation

This is a traditional cost minimization model, where the locations are called nodes, links between nodes are called edges, and olives flow from node to node.

Data

MaxSupply_n	Maximum supply at a node, kg	$\forall n \in N$
SupplyCost_n	Cost of supply at a node, \$/kg	$\forall n \in N$
Demand_n	Demand at a node, kg	$\forall n \in N$
Capacity_e	Capacity of link between nodes, kg sent	$\forall e \in E$
Distance_e	Distance between nodes, km	$\forall e \in E$
TransportRate_e	Cost of transport, \$/kg/km	$\forall e \in E$
Losses_e	Transit loss rate, kg/km	$\forall e \in E$

Sets

N	Nodes	$n = 1.. \text{Num Nodes}$
E	Edges	$e = 1.. \text{Num Edges}$

Model Formulation

Variables

$Supply_n$ = Supply at a node, kg $\forall n \in N$

$Flow_e$ = Flow between nodes, kg sent $\forall e \in E$

Model Formulation

Objective

$$\begin{aligned} \text{Minimize } Total \text{ cost} &= \sum_{n \in N} \text{SupplyCost}_n \times \text{Supply}_n \\ &+ \sum_{e \in E} \text{Distance}_e \times \text{TransportRate}_e \times \text{Flow}_e \end{aligned}$$

Minimize the cost by buying olives from suppliers and transporting the olives to our nodes.

Model Formulation

Subject to

$$Supply_n \leq MaxSupply_n \quad \forall n \in N$$

(c1) The supplier at each node has an upper bound on their capacity.

Model Formulation

Subject to

$$Supply_n \leq MaxSupply_n \quad \forall n \in N$$

$$Inflow_n = \sum_{e \in E} Flow_e \times (1 - Losses_e) \quad \text{if to node of } e = n$$

$$Outflow_n = \sum_{e \in E} Flow_e \quad \text{if from node of } e = n$$

$$Demand_n = Supply_n + Inflow_n - Outflow_n \quad \forall n \in N$$

(c2,3,4) The flow of olives at each node must balance.

Model Formulation

Subject to

$$Supply_n \leq MaxSupply_n \quad \forall n \in N$$

$$Inflow_n = \sum_{e \in E} Flow_e \times (1 - Losses_e) \quad \text{if to node of } e = n$$

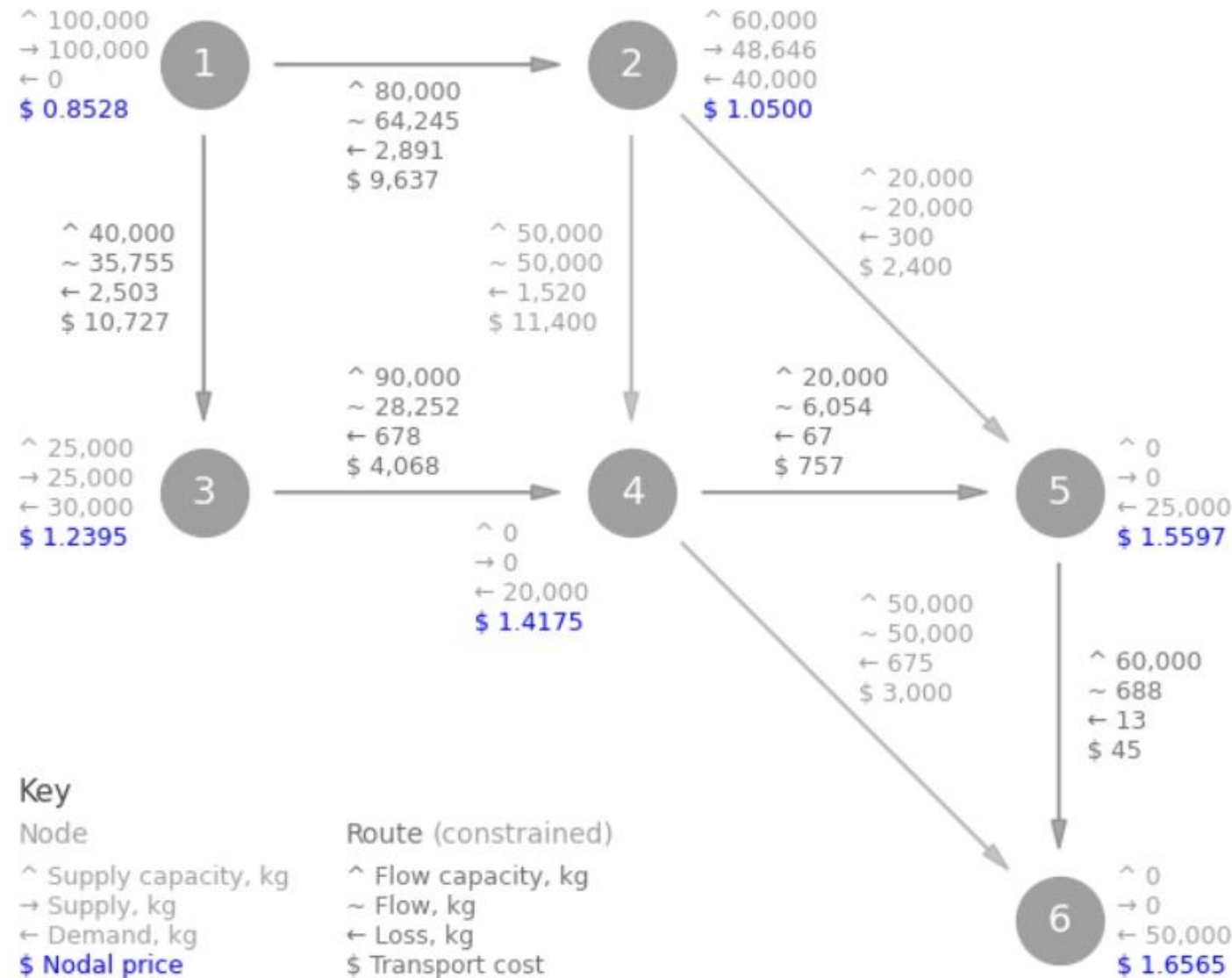
$$Outflow_n = \sum_{e \in E} Flow_e \quad \text{if from node of } e = n$$

$$Demand_n = Supply_n + Inflow_n - Outflow_n \quad \forall n \in N$$

$$Flow_e \leq Capacity_e \quad \forall e \in E$$

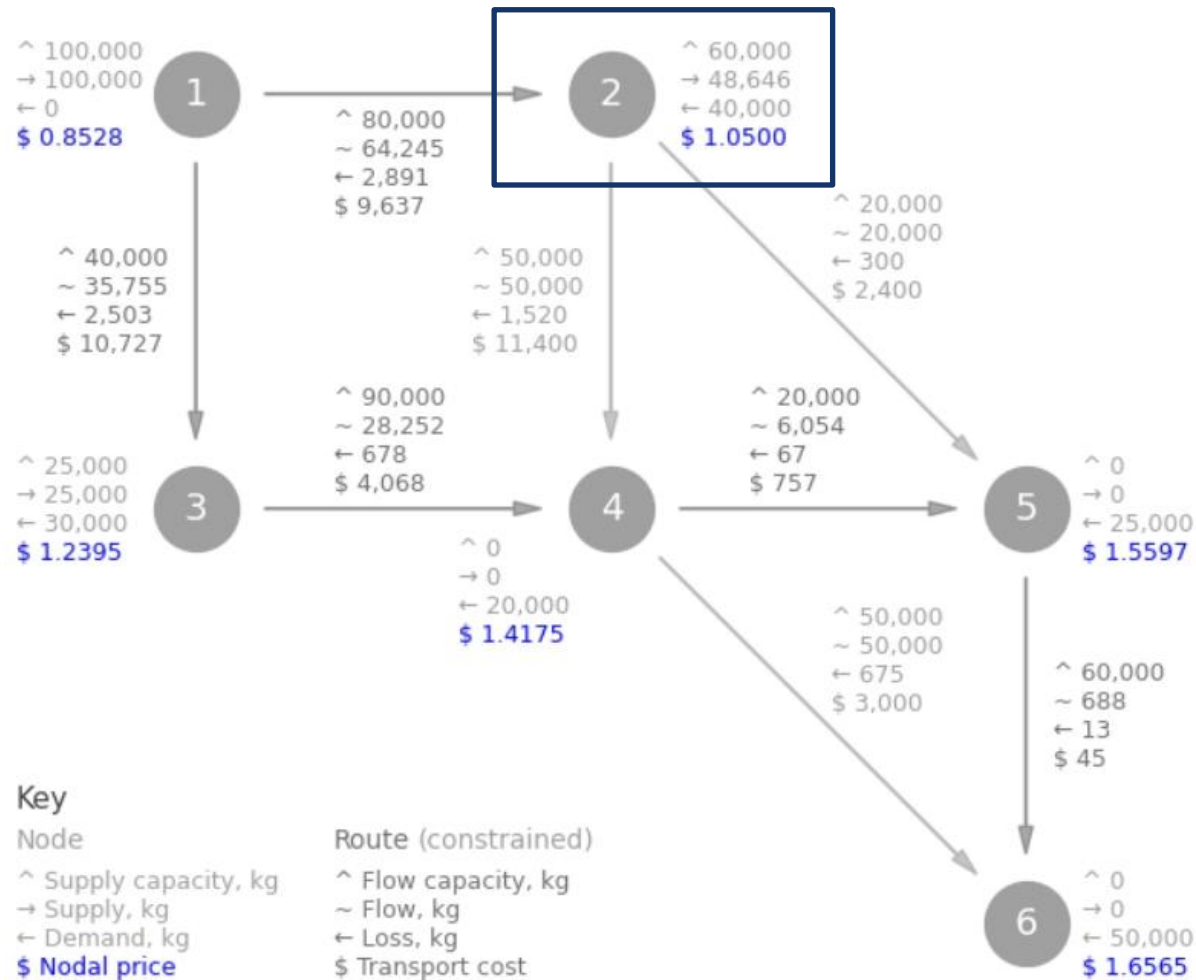
(c5) Each link has an upper bound on the transportation capacity.

Solution



So, what do the LMPs mean?

Solution



Node 2

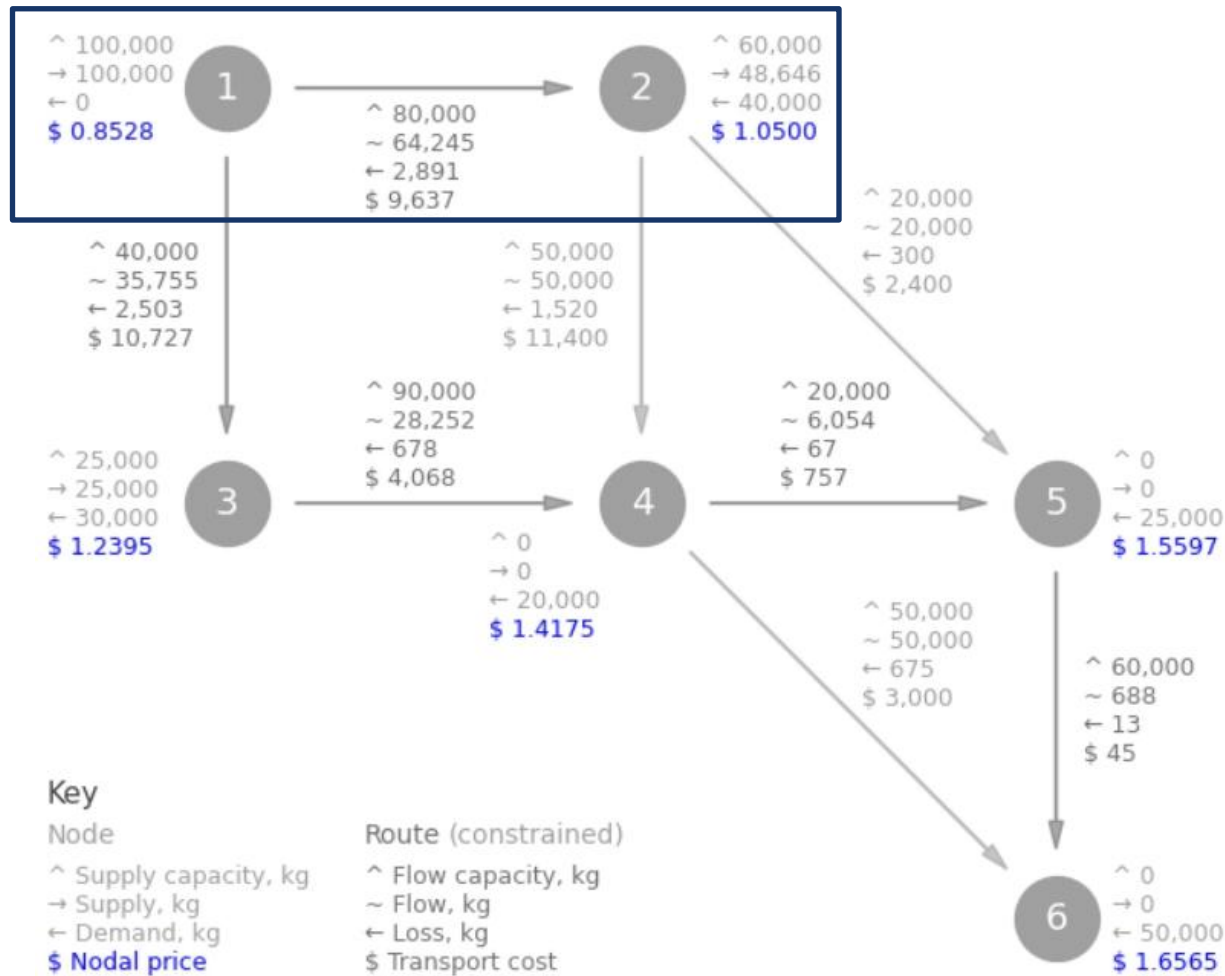
Current demand: 40,000 kg

If demand increases by a marginal amount of 1 kg, it would be supplied by the supplier at Node 2 as they have available capacity

Therefore, the marginal price of that extra 1 kg is the local supplier's cost of \$1.05/kg

Supplier	\$/kg	Max kg
Node 2	1.05	60,000

Solution

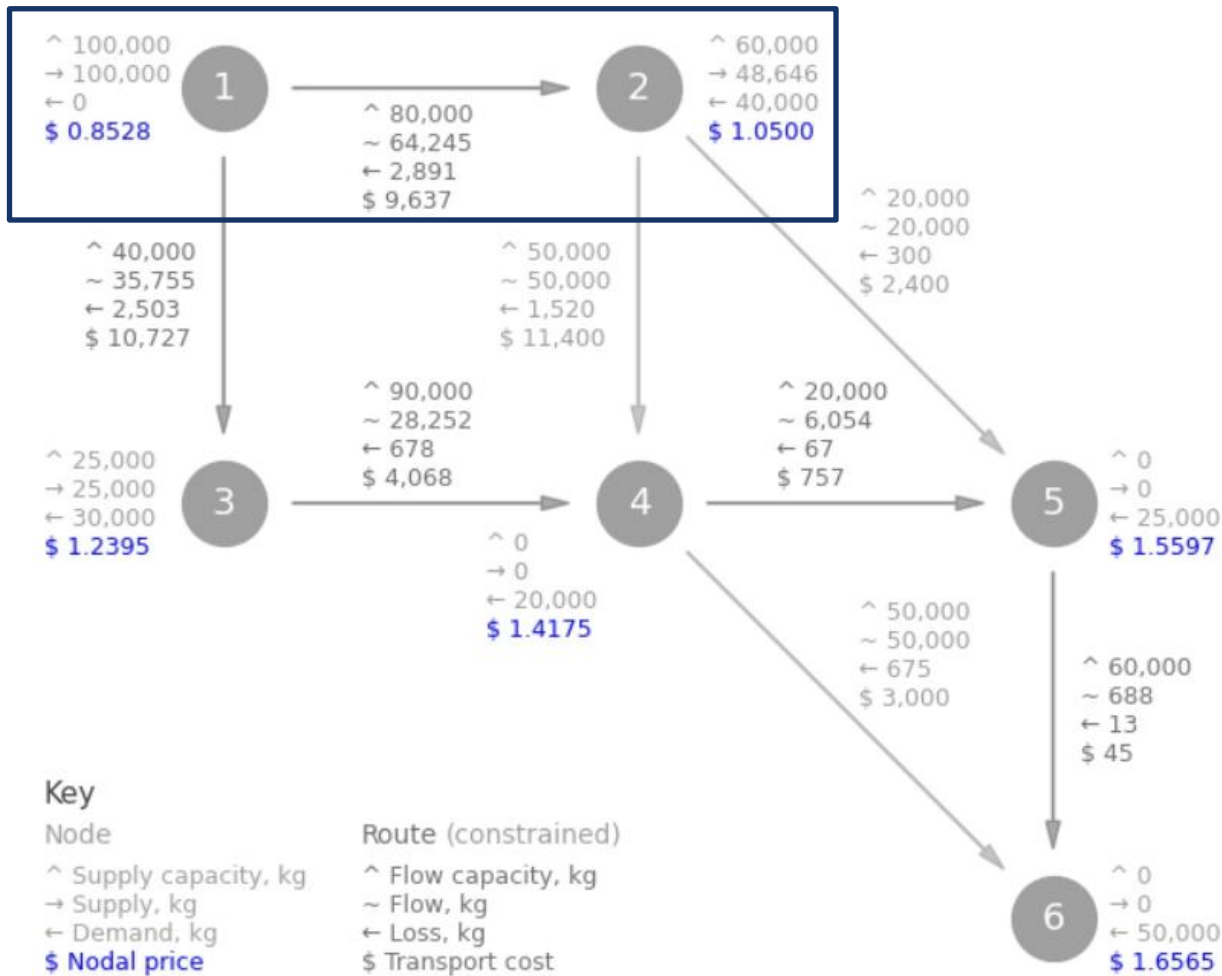


Node 1

If demand increases by a marginal amount of 1 kg, it cannot be supplied by the Supplier at Node 1 because they are already at their capacity

Instead, the Supplier at Node 2 supplies 0.9550 kg which increases costs by \$1.0028 (= 0.9550 * 1.05).

Solution

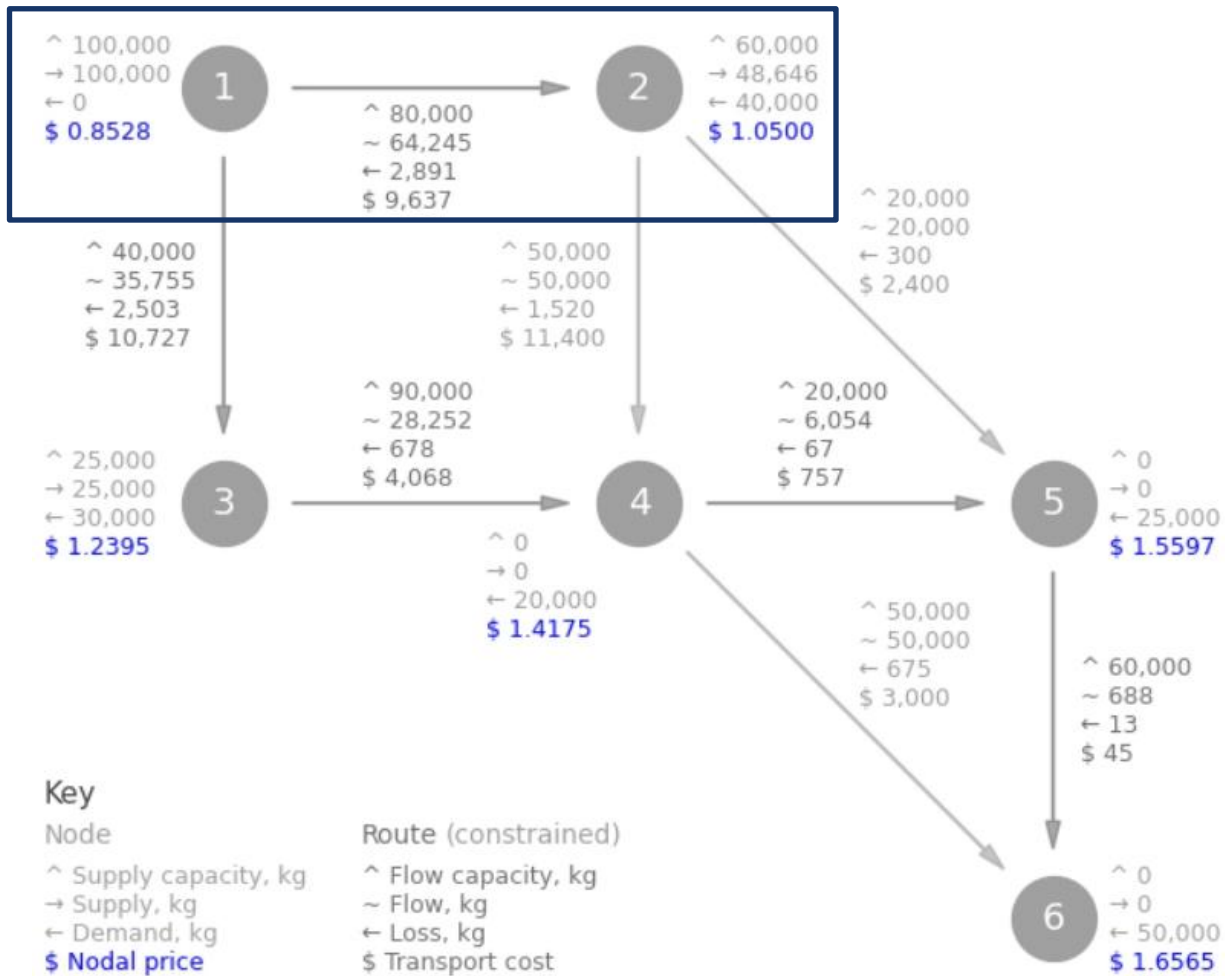


Node 1

Why don't Node 2 supply 1 kg?

The spoilage for 1 kg sent on the 150 km line between Node 1 and Node 2 is 4.5% (at a rate of 3.0%/100km), which is 0.0450 kg, so only 0.9550 kg extra supply is needed at Node 2 to deliver 1 kg extra at Node 1.

Solution



Node 1

The transportation cost also reduces, by \$0.1500 (= 1 kg * \$0.001/kg/km * 150 km).

Therefore, the net change in costs is
 $+\$1.0028 - \$0.1500 = \$0.8528$

How do we use the LMPs?

We tell our suppliers and contractors so that they can respond to the price signals!

? Question

What suggestions/requests you may have to our suppliers to improve overall efficiency of the system?

How do we use the LMPs?

We tell our suppliers and contractors so that they can respond to the price signals!

Scenario 1: Double transport capacity on constrained links

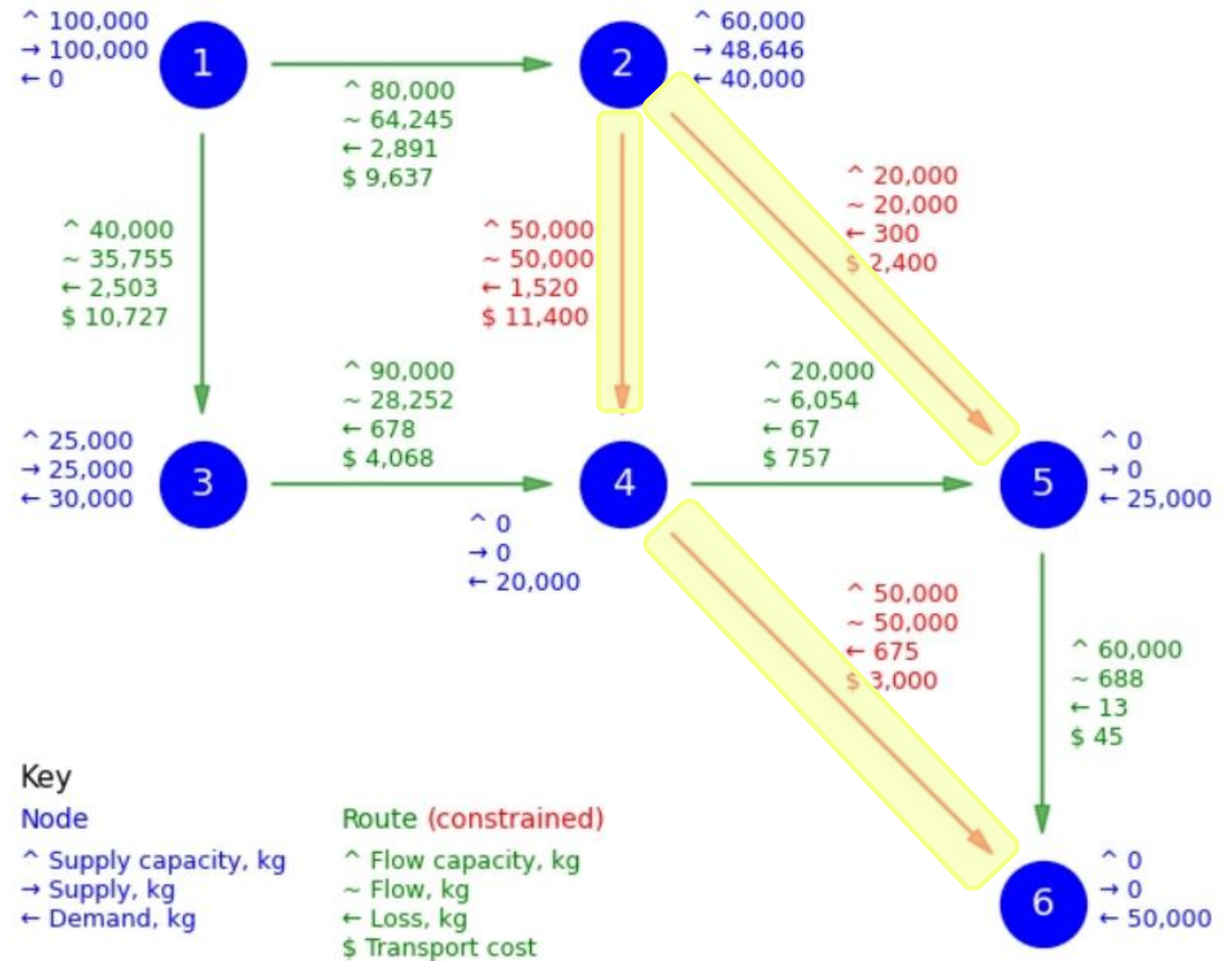
Scenario 2: Meet all demand from local suppliers

Scenario 3: Increase supply at Node 6

Scenario 4: Increase supply in the south and transport capacity to match

Scenario 1

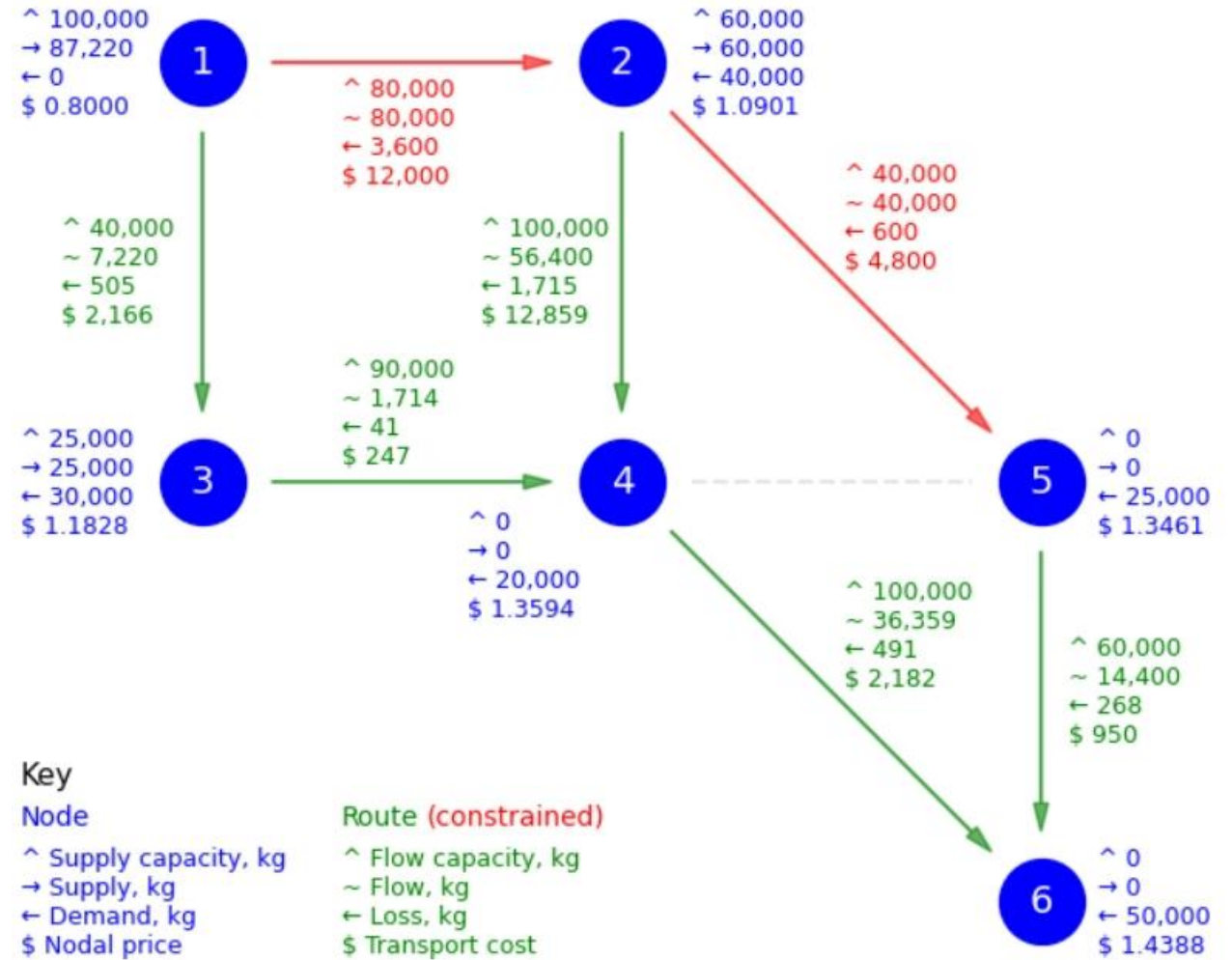
Double transport capacity on constrained links



Scenario 1 | Double transport capacity on constrained links

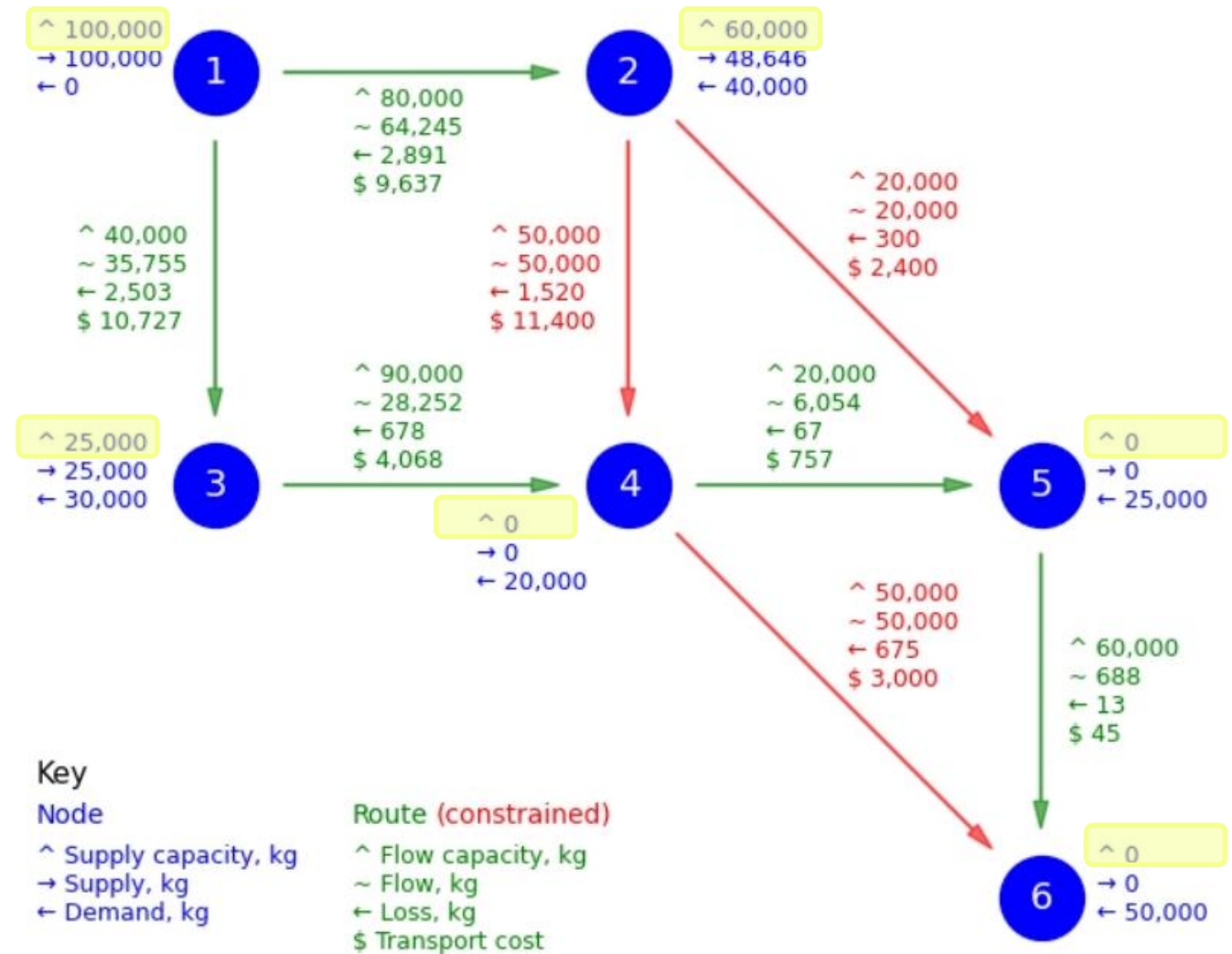
Better transportation reduces the direct transport costs and spoilage (so we need to buy less product)

Total Cost: \$190,479.74



Scenario 2

Meet all demand from local suppliers

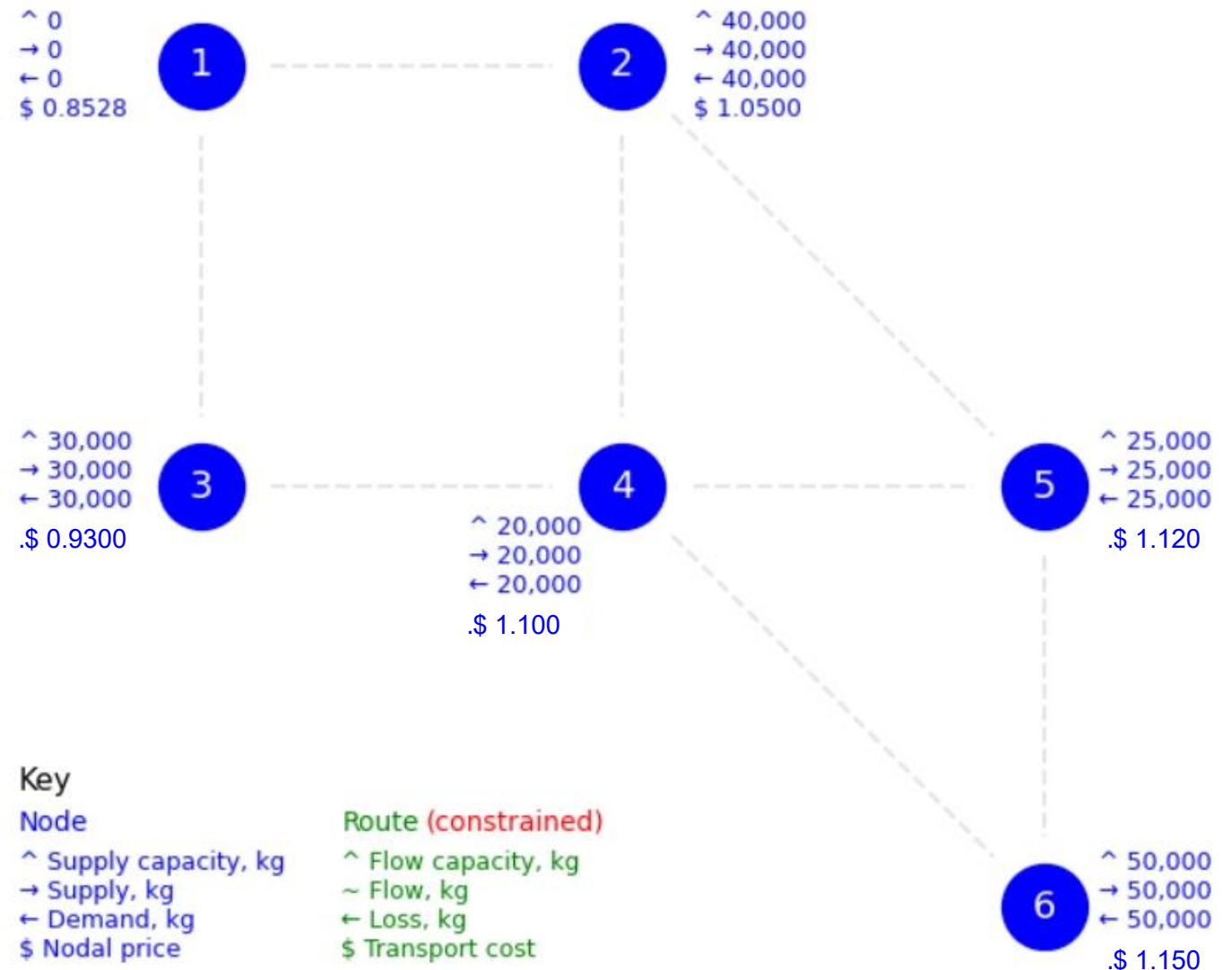


Scenario 2 | Meet all demand from local suppliers

Total Cost: \$176,500

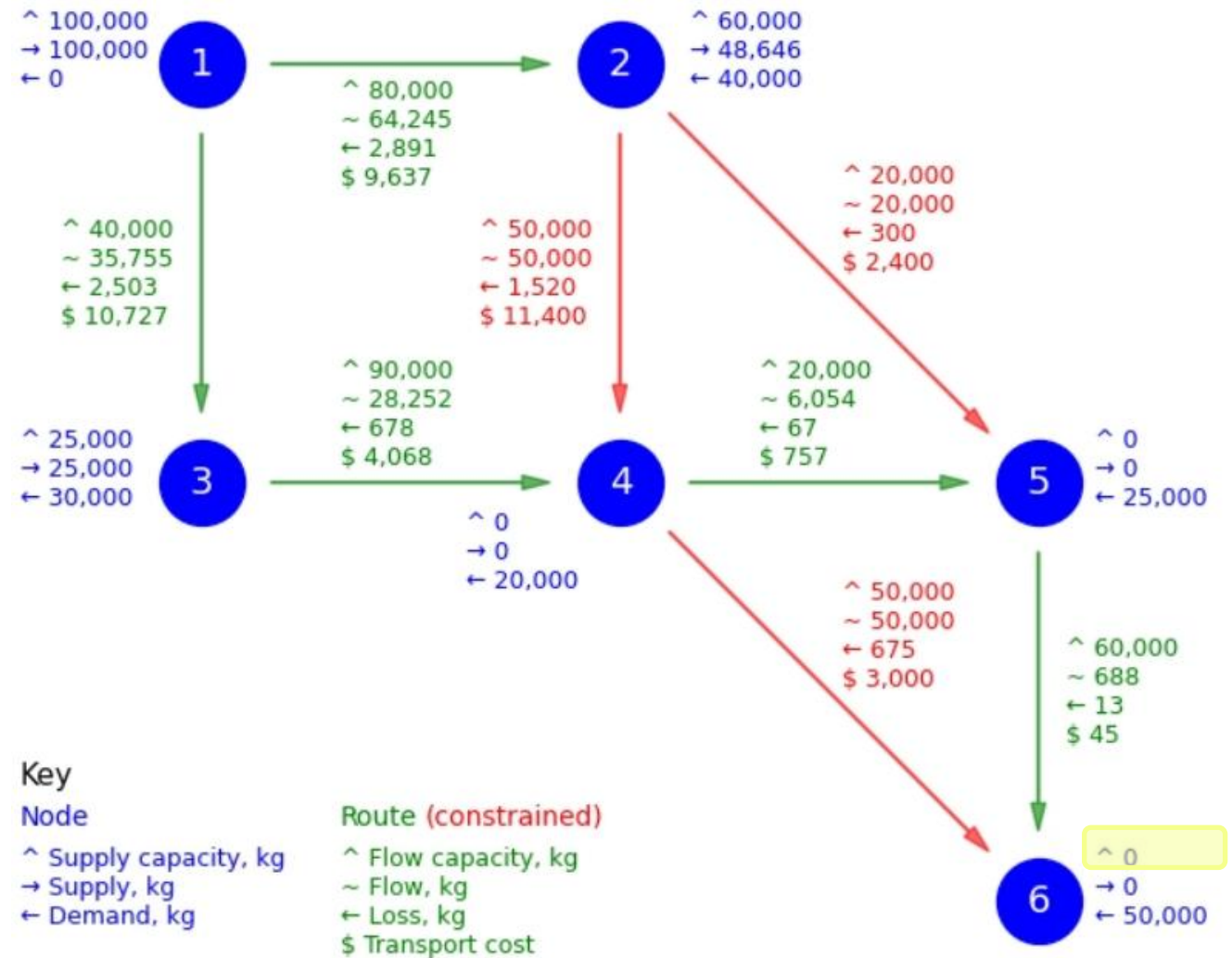
We pay more for the product but eliminate the transportation cost (including spoilage) entirely.

This result is cheaper than Scenario 1, but what if this is not possible?



Scenario 3

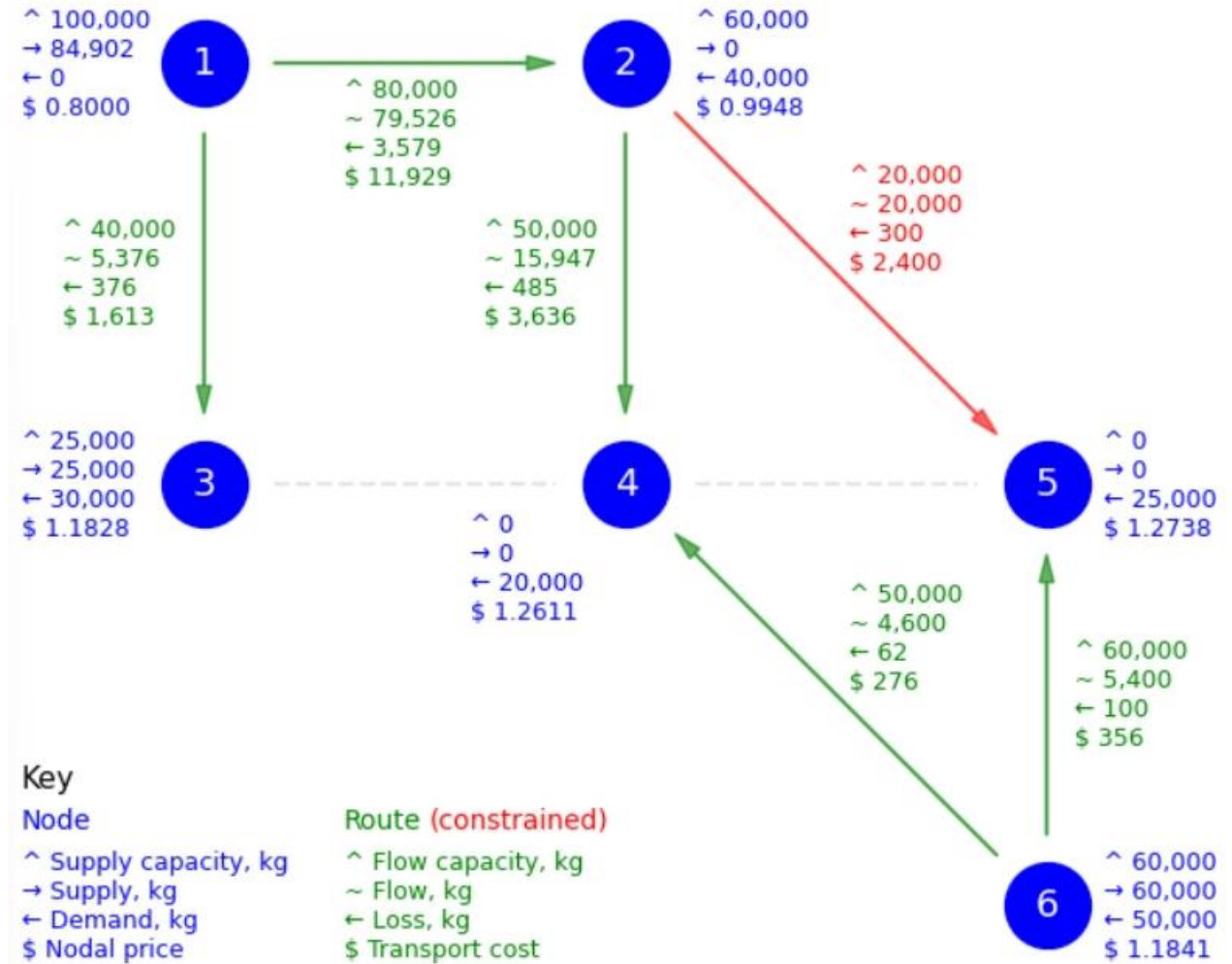
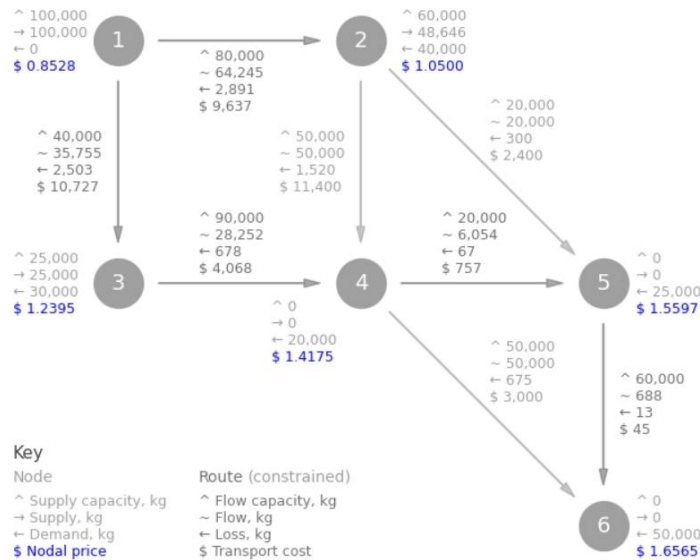
Increase supply at Node 6



Scenario 3 | Increase supply at Node 6

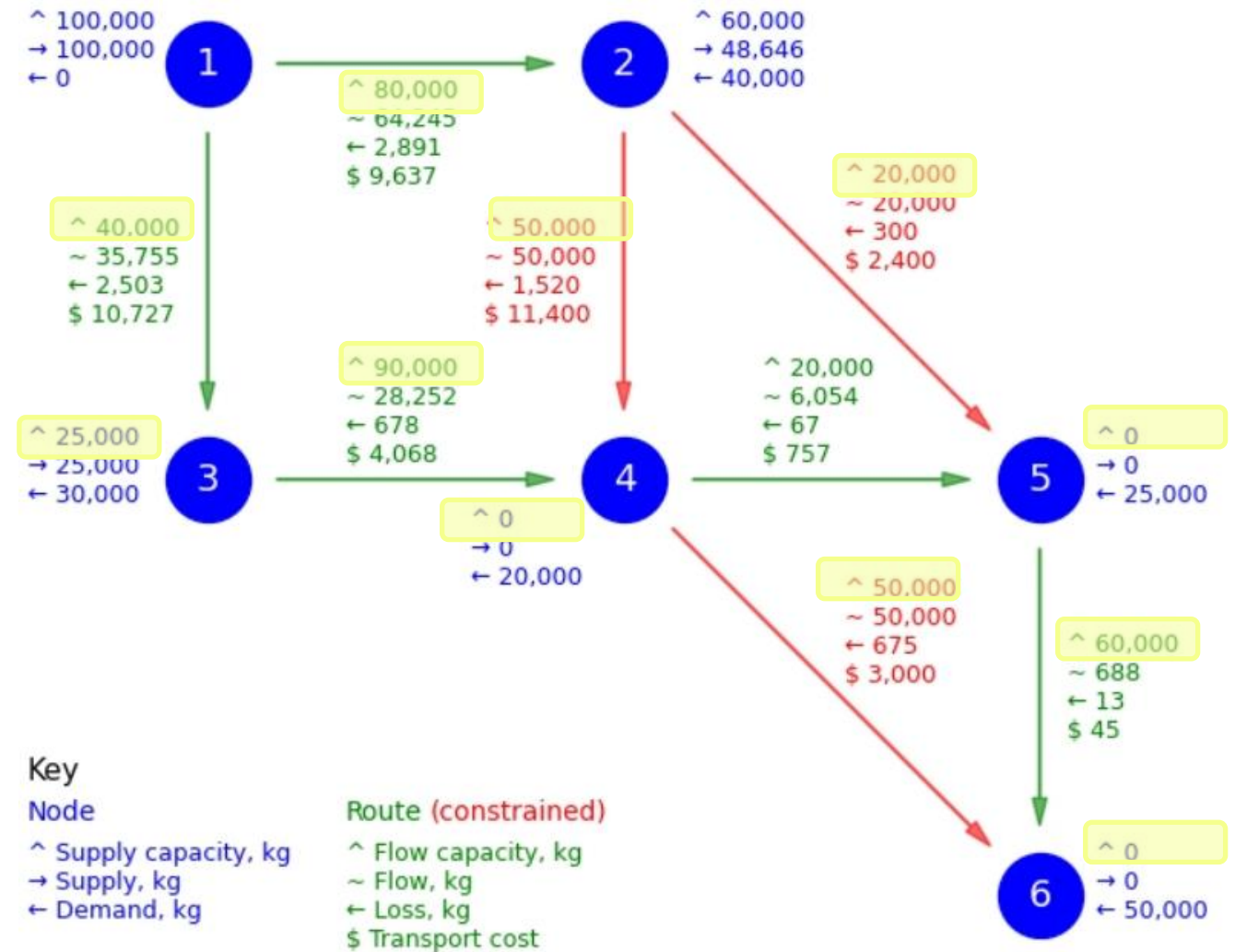
Total Cost: \$ 179,632

All the LMPs reduce, some of them substantially!



Scenario 4

Increase supply in the south and transport capacity to match

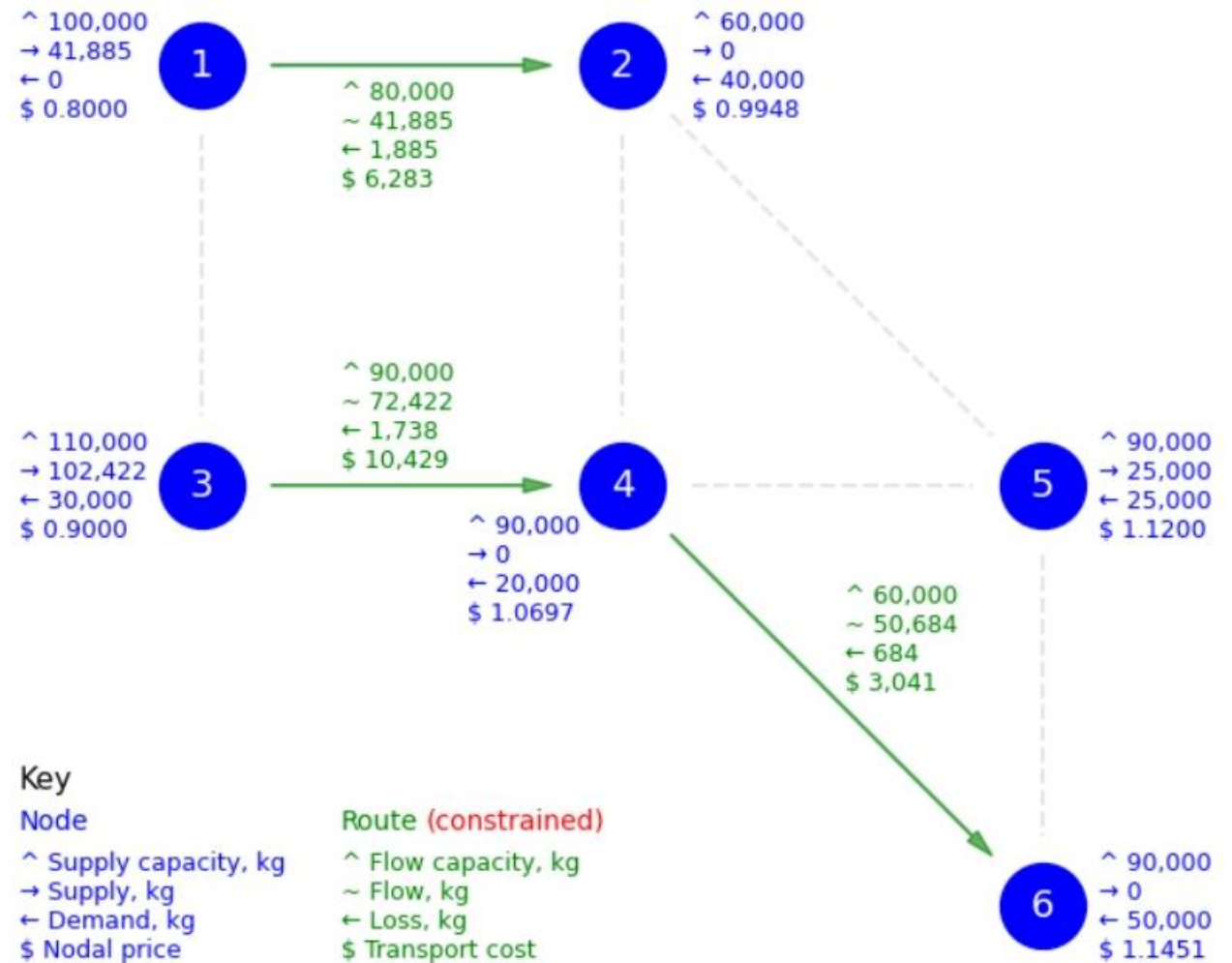


Scenario 4 | Increase supply and transport capacity to match

- **Total Cost: \$ 173,440**

This result shows that transporting olives around the network is cheaper than all nodes being supplied locally.

In other words, there are gains from trade between the nodes, even though trade incurs transportation costs.

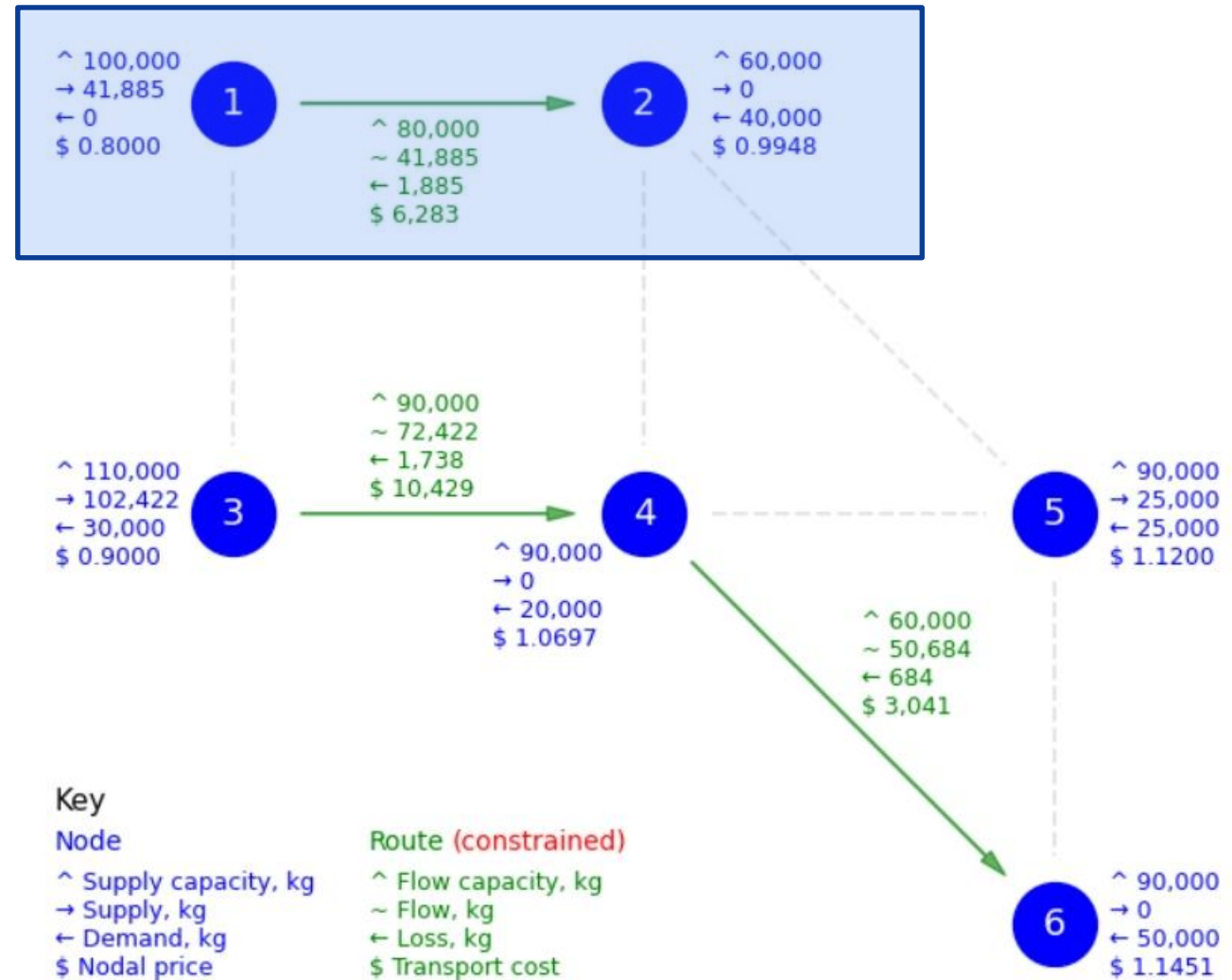


Scenario 4 | Increase supply and transport capacity to match

● Total Cost: \$ 173,440

The supplier at Node 1 is the cheapest on the island.

In this scenario, they supply all demand at Node 1 and Node 2. Allowing for transport and spoilage, it is cheaper for the supplier at Node 1 to supply our demand at Node 2 rather than using the Node 2 supplier directly (Node 2's LMP is \$0.9948/kg compared with Node 2 supplier cost of \$1.0500/kg).



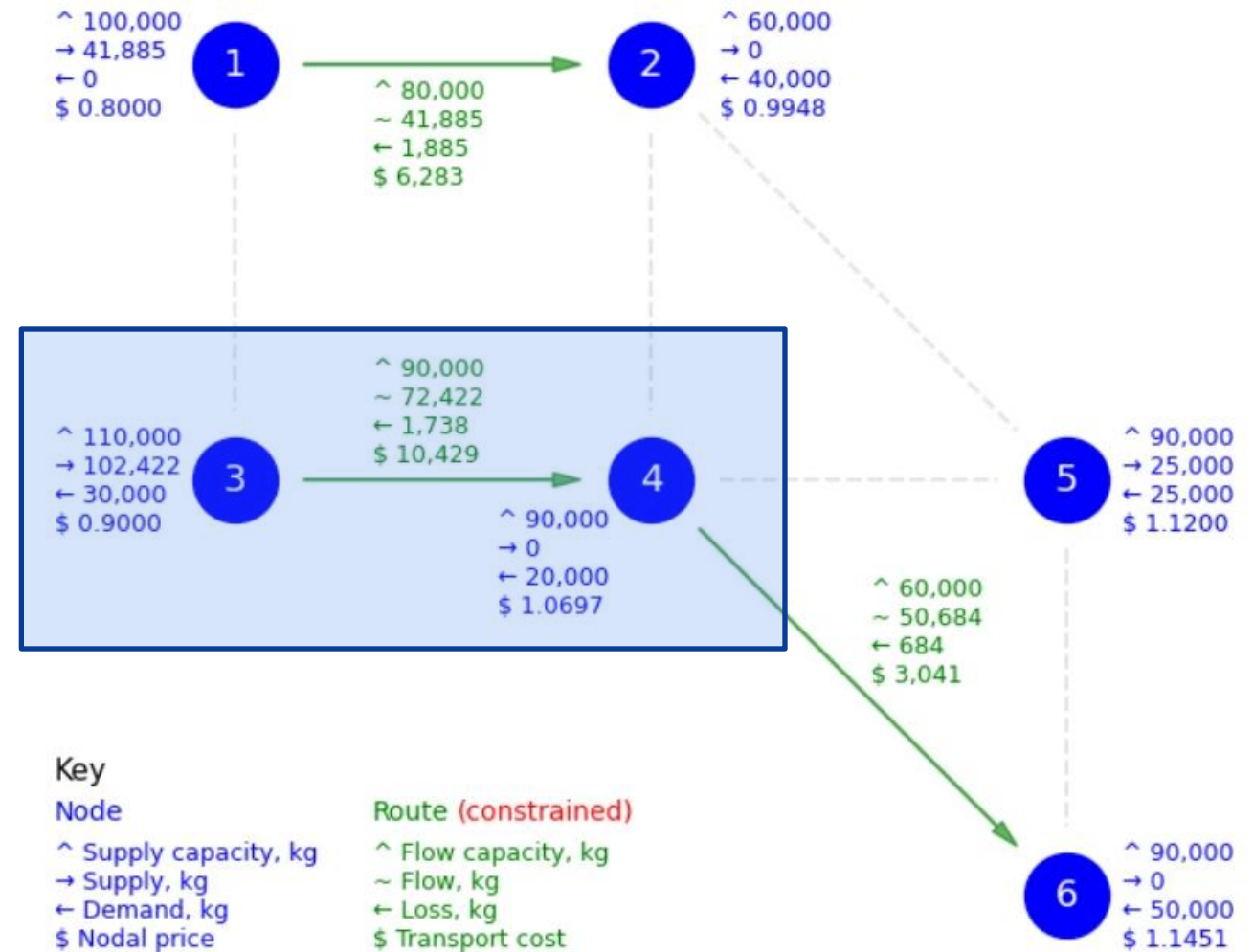
Scenario 4 | Increase supply and transport capacity to match

● Total Cost: \$ 173,440

Supply at Node 3 is fairly cheap, at \$0.9000/kg, so their volume increases substantially.

They supply all the demand at Nodes 3, 4, and 6.

- The LMP at Node 6 is \$1.1451/kg, indicating that supply from Node 3 plus transport and spoilage is slightly cheaper than local supply at Node 6 given the Node 6's supply cost of \$1.1500/kg.
- Similarly for Node 4. Therefore, we have no local supply at Nodes 4 and 6.

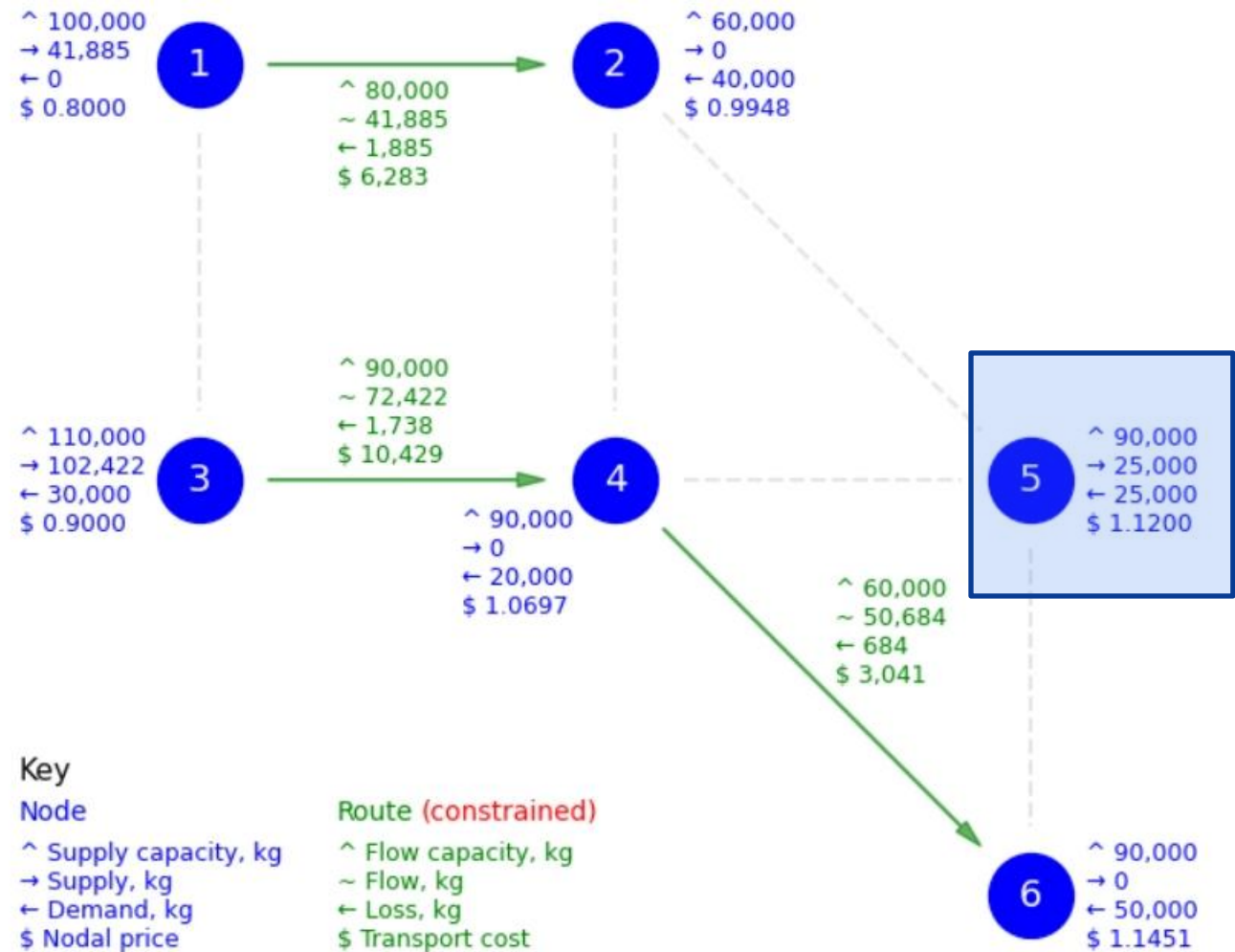


Scenario 4 | Increase supply and transport capacity to match

- **Total Cost: \$ 173,440**

Transport to Node 5 is expensive.

So, the cheapest option is for the demand at Node 5 to be met entirely from the local supplier at Node 5.



Summary of Scenarios & Conclusions

 **Markets are a wonderful thing!** 

In this case, we applied LMP to the supply of olives to restaurants.

The goal is to provide the restaurants' suppliers and contractors with price signals to incentivize efficient behavior.

After discussing with your classmates, could you share with the class what your take-aways are from this activity?